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The Real Effects of Mandatory Corporate Social Responsibility Reporting in China

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We investigate the real effects of corporate social responsibility (CSR) reporting regulation enacted in China in 2008 covering both state-owned enterprises (SOEs) and non-state-owned enterprises (NSOEs). As the CSR reporting requirement applies only to select firms, the regulation presents a unique setting that provides us with a suitable control sample of firms not subject to the regulation. Using a difference-in-differences approach, we document that relative to firms that are not affected by the regulation, NSOEs and SOEs with low state ownership subject to the mandate are associated with increases in CSR expenditures. Moreover, the real effects manifest in non-core initiatives for these firms, whereas they manifest in core operating activities for SOEs with high state ownership. Finally, the effects of the mandatory CSR reporting on profitability and shareholder value are negative.

Key words: mandatory disclosure; real effects; corporate social responsibility; non-state-owned enterprises; non-core activities
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1. Introduction

We investigate the differential real effects of the 2008 corporate social responsibility regulation in China (hereafter, the CSR regulation or mandate) on state-owned enterprises (SOEs) and non-state-owned enterprises (NSOEs). Although China has made tremendous strides in recent times as a key player in the global arena and has become a "factory of the world," both SOEs and NSOEs have been known to be socially negligent in the eyes of the public (Young 2002). The CSR regulation is just a disclosure requirement—it does not mandate spending on CSR activities (unlike in India).[1] Would a mere disclosure mandate induce firms to increase CSR initiatives?

SOEs and NSOEs are markedly different in terms of corporate governance, incentive structures, institutional norms, and customs. The government's stake in NSOEs is minimal (with a mean of 4.4%), and therefore NSOEs are primarily shareholder value-maximizing firms. Moreover, publicly traded SOEs are subject to the type II agency conflict between the state and minority shareholder blocs. Thus, SOEs and NSOEs face vastly different institutional pressures that have considerable impact on decision making (Balakrishnan et al. 2010, Goodstein 1994, North 1990, Zucker 1987).

We focus on the impact of the CSR regulation on NSOEs in comparison with SOEs because NSOEs play a significant role in the Chinese economy. In a speech to representatives of entrepreneurs in 2018 (CGTN 2018), President Xi Jinping observed that China had more than 27 million private companies accounting for \$24 trillion in registered capital, and that these companies contribute more than 50% of tax revenue, 60% of GDP, and almost 80% of urban employment. The median market capitalization of NSOEs in our sample is \$413 million, which is 74% of the median market capitalization of SOEs. Examining the impact of the CSR regulation on NSOEs, therefore, appears paramount.[2]

It is not immediately clear whether the CSR mandate will induce NSOEs to increase investments in CSR initiatives. On the one hand, being just a disclosure mandate, it does not compel NSOEs to take actions at shareholders' expense. On the other hand, the CSR disclosure mandate heightens global attention and scrutiny (He et al. 2012). If these mandated disclosures feed the perception that not enough is being done, the consequent backlash (in the form of reputation loss, loss in global market share, punitive damages, and other indirect penalties) could be harmful to shareholders' interests. Thus, the regulation presents NSOEs with a trade-off. To the extent this trade-off forces some NSOEs to undertake CSR initiatives they would have not taken absent the disclosure mandate, we should expect to see the disclosure regulation triggering real effects in NSOEs with consequent implications for shareholder value.

In related work, Chen et al. (2018) provide evidence that following the regulation, profitability declines for SOEs but not for NSOEs, but pollution levels decrease in "low-SOE" cities, that is, in cities with more NSOEs. The results relating to NSOEs are especially striking because they imply that NSOEs increase pollution-reducing expenditures following the CSR mandate but their profitability does not decline. A question that arises naturally is: If NSOEs can implement socially beneficial CSR activities without hurting their own profits, would NSOEs not have done so (*and* disclosed such activities) without a government mandate? Moreover, it would seem that NSOEs are less likely to spend on CSR initiatives that could result in a wealth transfer away from shareholders (Moser and Martin 2012), in which case a government mandate appears necessary to trigger socially beneficial wealth transfers. Moreover, we would expect such regulation-induced CSR expenditures to adversely affect profitability and shareholder value.

Motivated by these considerations, we seek direct evidence on whether the CSR regulation affects the level of CSR expenditures incurred by NSOEs relative to SOEs by hand-collecting data on these expenditures before the mandate and comparing them with CSR expenditures they report after the mandate. We analyze the impact of the CSR regulation on profitability immediately after the regulation (years 2008 and 2009) and in later years (years 2010, 2011, and 2012) to allow for the possibility that CSR expenditures can affect profits with a lag.[3] We also evaluate the impact of the CSR regulation on shareholder value.

SOEs, unlike NSOEs, are subject to type II agency conflicts between state priorities and minority shareholder interests. Specifically, SOEs with high state ownership are more likely to align their CSR actions in line with state priorities. Moreover, we would expect SOEs with low state ownership to be less encumbered in pursuing the interests of shareholders (i.e., be less subject to the type II agency problem). Accordingly, we analyze the effect of the extent of state ownership in our analysis of the impact of the CSR regulation.

A challenge faced by researchers in providing evidence on the real effects of disclosure regulations is the identification of a control sample unaffected by these regulations (Leuz and Wysocki 2016). In this respect, the Chinese CSR regulation presents a unique setting to investigate the causal effects of disclosure because it only applies to select firms (hereafter, *treatment* firms). Therefore, firms not subject to the mandate constitute a control sample (hereafter, *control* firms). Consequently, we employ a difference-in-differences approach to identify the real effects of the CSR regulation.

We find that both SOE and NSOE treatment firms increase their CSR expenditures following the regulation relative to control firms.[4] When we control for systematic pre-regulation differences between treatment and control firms (via a "parallel trends" test), we continue to find that NSOE treatment firms increase their CSR expenditures following the regulation. However, only SOE treatment firms with *low* state ownership increase their CSR expenditures which is consistent with the notion that these firms are similar to NSOEs in terms of shareholder value-maximizing incentives. We do not find a similar trend for SOE treatment firms with *high* state ownership.

Interestingly, our results show that core operating costs (cost of goods sold and total operating expenses) increase markedly for SOE treatment firms in the post-regulation period (consistent with Chen et al. 2018), but *not* for NSOEs. Moreover, we find that the results for SOE treatment firms are driven solely by those with high state ownership—there is no increase in core operating costs for SOE treatment firms with low state ownership. However, we find treatment NSOEs and SOEs with low state ownership engage in *non-core* CSR activities—CSR special items increase significantly for these firms post-regulation.[5] In contrast, we do not find evidence of an increase in CSR special items for treatment SOEs with high state ownership. These results support the narrative that shareholder-value maximizing firms prefer *non-core* CSR activities to core ones given that non-core items are considered less value-relevant and likely discounted by the market. Additionally, accounting standards allow for the capitalization of certain CSR expenditures relating to employee protection and environmental protection. An analysis of these expenditures indicates that they increase significantly for NSOE treatment firms following the regulation (after we control for pre-regulation trends). Taken together, these results indicate that the CSR regulation has a significant impact on NSOEs by inducing them to devote some resources toward CSR initiatives that they previously did not.

Our results indicate a decline in profitability for both SOE and NSOE treatment firms following the regulation, whereas Chen et al. (2018) find that profitability decreases primarily for SOEs. Furthermore, we find that this profitability decline occurs for SOEs and NSOEs only after a lag (in the period of 2010 to 2012). We also find that price-to-book ratios decline markedly for both SOE and NSOE treatment firms from pre- to post-regulation period relative to the control firms. Specifically, for NSOEs, the negative valuation impact is consistent with the market interpreting the real effects of the CSR regulation as being detrimental to shareholder value. In other words, the increased global scrutiny that the CSR regulation brings imposes a cost on shareholder value-maximizing firms by forcing them to boost CSR spending.

In additional analysis, we also explore whether the CSR regulation has real effects with respect to credit flows in value chains. The CSR reporting guidelines of Shenzhen Stock Exchange (SZSE 2008) require firms subject to the CSR mandate to disclose measures taken toward "protection of suppliers, customers, and consumers," which brings about more intensive attention and scrutiny on value chain relationships.[6] Indeed, our empirical analysis reveals that following the regulation, NSOEs increase their credit provision to customers, whereas SOEs in general turn over their accounts payable more quickly. In other words, the real effects of the CSR regulation appear to extend to value chain relationships.

A strand of empirical research documents real effects of specific mandatory disclosures (see Roychowdhury et al. 2019 for a review). As the literature suggests, disclosures can have real effects by reducing information asymmetry, improving monitoring, and mitigating financial constraints. Chuk (2013) shows that mandatory disclosure of pension asset composition induces changes in pension asset allocation. Hayes et al. (2012) show that mandatory disclosure of stock options expense induces changes in option use. Cho (2015) shows that mandatory segment reporting induces changes in internal capital allocation across segments. Dyreng et al. (2015) show that mandatory subsidiary disclosure induces changes in tax avoidance. Cheng et al. (2013) show that mandatory disclosure of internal control weaknesses induces changes in corporate investment efficiency.

A unique aspect of China's 2008 CSR regulation is that it does not mandate spending on CSR activities. In contrast, the Indian regulation mandates CSR spending directly on firms that come under the purview of its mandate. Manchiraju and Rajgopal (2017) analyze shareholder wealth effects of the Indian regulation and document a decline in stock price for firms that were forced to spend additional resources on CSR. Our focus is on whether the CSR disclosure mandate of China triggers real effects in the form of new CSR initiatives—especially in NSOEs—despite there being no explicit mandate requiring them to do so. In this respect, our paper is similar in spirit to Christensen et al. (2017), who find that dissemination of corporate mine safety records through financial reporting triggers real effects, although such safety records are already publicly available elsewhere.

We make two contributions to the literature. First, we hypothesize and document that the CSR disclosure regulation has real effects on NSOEs. Our results thus provide additional insight by documenting a broader scope of the real effects beyond the real effects on SOEs documented in Chen et al. (2018). Second, we document that the real effects of the CSR regulation take place through different channels for SOEs and NSOEs. For *SOEs with high state ownership*, the real effects take place in the form of *core* operating activities. However, for *NSOEs* and *SOEs with low state ownership*, they take place through *non-core* initiatives. Because government-run SOEs with high state ownership are supposed to be mindful of social responsibilities, their CSR investments affect core, sustainable operating expenses. Moreover, NSOEs and SOEs with low state ownership are more inclined to act in shareholders' interests, and they respond to the CSR regulation in ways that do not affect their core operations and they treat CSR initiatives as noncore. Nevertheless, shareholders appear to view these initiatives as value-decreasing.

The paper proceeds as follows. Section 2 develops the hypotheses. Section 3 presents sample selection and empirical methodology. Section 4 presents the main results. Section 5 adds additional analyses. Section 6 concludes the paper.

2. Hypotheses

While the Chinese CSR regulation mandates disclosure, there is nevertheless a *voluntary* aspect to this regulation, which is that there is no imposition on firms to engage in CSR activities. Some firms, especially NSOEs, may find it beneficial to engage in certain CSR activities because they advance shareholder value. Even in the absence of a CSR disclosure mandate, we would expect these firms to disclose their CSR activities in their financial statements and notes because no disclosure costs exist. In particular, the unraveling result of Grossman (1981) and Milgrom (1981) would apply—non-disclosure brings no benefit to these firms. For other NSOEs, CSR initiatives are potentially deterred by a wealth transfer from shareholders to other stakeholders (Moser and Martin 2012). Absent any CSR initiatives, these firms have nothing to disclose. By the same token, if the regulation triggers new CSR initiatives, these firms have nothing to gain by *not* disclosing—whether such disclosures are mandated or not. In brief, in the full disclosure equilibrium, any firm engaging in CSR activities—regardless of whether these activities advance or hurt shareholder value—will always disclose because non-disclosure only imposes a cost but brings no benefit.

Thus, in developing our hypotheses, we maintain the premise that firms will fully disclose any CSR activities they engage in (voluntarily or otherwise). The question is whether the disclosure *mandate* forces firms to engage in *additional* CSR initiatives. To the extent mandated stand-alone CSR reports expose firms to *heightened* global attention and scrutiny, disclosure of *inadequate* CSR activities imposes a cost—a cost associated with adverse consequences such as reputation loss, loss in global market share, punitive damages, and other indirect penalties. For example, if a petrochemical company had inadequate pollution reduction or environmental protection programs, public awareness of that fact could potentially be damaging to shareholder value. Thus, these firms face a trade-off: Either maintain the current level of CSR activities to prevent a wealth transfer from shareholders to other stakeholders or engage in *additional* CSR activities (and disclose them) at a cost to shareholders.

It seems reasonable to expect that the average NSOE firm will be affected by additional scrutiny the mandated CSR reporting would bring because at least some of these firms are likely to be perceived as having inadequate CSR initiatives. In other words, it may be beneficial for shareholder value-maximizing firms to undertake additional CSR initiatives they would not have otherwise taken—"doing well by doing good" as it were. These real effects can take the form of (1) direct CSR expenditures, (2) increases in cost of goods sold and total operating expenses, and (3) increases in non-core special items. We first examine the impact of the CSR regulation on direct CSR expenditures on NSOEs by testing the following hypothesis:

Hypothesis H1. *CSR expenditures increase post-regulation for NSOEs subject to the CSR regulation.*

In contrast, SOEs' operations are subject to the government's priorities and, therefore, we would expect them to heed social welfare with or without regulation. When viewed from this perspective, the CSR disclosure mandate should have no real implications for SOEs. Yet, Chen et al. (2018) provide evidence that the mandate has been effective in changing the behavior of SOEs. To provide additional insight in this respect, we partition the SOE subsample based on the extent of ownership. By design, the state has a controlling interest in SOEs with high state ownership. This gives rise to a more intensive conflict between the majority shareholders and minority shareholders and imposes type II agency costs on minority shareholders. Thus, under the premise that government ownership increases pressure to engage in CSR activities even absent CSR mandate, the actions of SOEs with high state ownership are more likely in line with the state's priorities. Consequently, for these firms, we should not expect any boost in CSR expenditures post-regulation, relatively speaking. In contrast, SOEs with low state ownership may be viewed as being more like NSOEs in terms of their disposition to the CSR regulation (lower type II agency costs). Therefore, we expect that the CSR reporting mandate will impact SOEs with low state ownership in much the same way as NSOEs. Accordingly, we test the following hypothesis:

Hypothesis H2. *CSR expenditures are greater for SOEs with low state ownership relative to SOEs with high state ownership subject to the CSR regulation.*

We next consider the impact of CSR mandatory disclosure on operating costs. According to the guidelines on mandatory CSR reporting in China (SSE 2008, SZSE 2008), CSR activities are classified into five categories: shareholder and creditor protection, employee protection, supplier and customer protection, environmental protection, and public relations. Thus, some of these activities will impact core operating expenses, and others will impact non-core special items.

We turn first to *core* operating expenses. Manufacturing companies account for 56% of our sample, and cost of goods sold is the biggest expense item, comprising about 75% of sales on average. CSR include employee protection and environmental protection. For example, many CSR programs include investments in work-life balance, health and safety, and employee involvement (Flammer and Luo 2017). Barnett and Salomon (2012) claim that firms with higher corporate social performance incur higher costs such as investing in additional employee benefits and pollution reduction.

Chen et al. (2018) find that core operating expenses increase for firms subject to the CSR regulation, but do not examine the effects for SOEs and NSOEs separately. Our interest lies in examining whether operating costs, core or non-core, increase for firms that are driven by shareholder-value maximization, that is, NSOEs and SOEs with low state ownership. We use two alternate measures of core operating expenses—total operating expenses computed as the sum of cost of goods sold and SG&A expenses, and cost of goods sold alone.

Hypothesis H3A. *Cost of goods sold-to-sales ratio and total operating expenses-to-sales ratio increase for NSOEs and SOEs with low state ownership subject to the CSR regulation.*

Many CSR expenditures involve non-core activities such as donations to various community initiatives, pollution control, and waste recycling. These expenditures may be expensed as below-the-line non-core items, or capitalized delaying the income statement impact of these items. It is possible that NSOEs and SOEs with low state ownership increase these types of CSR activities rather than core operating activities that adversely affect core operating margins.[7] Evidence suggests that investors value core and non-core items in the income statement differently, and that managers manage non-core items to smooth income before special items (Ronen and Sadan 1981) and to smooth core operating income (Barnea et al. 1976). Mcvay (2006) provides evidence that managers misclassify core operating expenses to special items to overstate core earnings to meet or beat analysts' forecasts. Kinney and Trezevant (1997) show that managers have a greater incentive to reveal the transitory nature of income-decreasing special items.

Based on these studies, we hypothesize that NSOEs and SOEs with low state ownership have more pronounced incentives to classify CSR expenditures as non-core items and even capitalize some of them to underplay any negative implications of CSR expenditures on shareholder value. For a subsample of these firms, we are able to manually identify CSR-related special items before and after the regulation and those that are capitalized, and therefore we test the following hypothesis.

Hypothesis H3B. *Non-core expenses-to-sales ratio and CSR special items-to-sales ratio increase for NSOEs and SOEs with low state ownership subject to the CSR regulation.*

Hypothesis H3C. *Capitalized CSR special items-to-assets ratio increases for NSOEs and SOEs with low state ownership subject to the CSR regulation.*

Next, we examine the impact of the CSR regulation on profitability and shareholder value. We view regulation as imposing an additional constraint on firms in maximizing shareholder value, especially in the case of NSOEs. Moreover, it is plausible that shareholders ultimately benefit in the long run if their firms act in a socially responsible way. This "strategic CSR" view advances the notion that socially responsible actions build trust and reputation among all stakeholders, making companies stronger and more enduring, thus enhancing shareholder value (Godfrey et al. 2009, Wang et al. 2014). However, any shareholder value-maximizing firm should be expected to engage in such value-maximizing CSR activities in the normal course of business anyway—it should not take a CSR disclosure mandate to trigger such actions. Consequently, we posit that the CSR disclosure mandate would decrease profitability and shareholder value for NSOEs and SOEs with low state ownership.

Hypothesis H4A. *Profitability ratios (ROA and ROE) decrease post-regulation for NSOEs and SOEs with low state ownership subject to the CSR regulation.*

Hypothesis H4B. *Price-to-book ratios decrease for NSOEs and SOEs with low state ownership subject to the CSR regulation.*

We note that Chen et al. (2018) do not observe any profitability decline for NSOEs. To gain additional insight in this regard, we evaluate profitability decline immediately following the regulation (2008 and 2009) and in the longer term (from 2010 to 2012).

3. Sample and Methodology

Our initial sample consists of all firms listed on the Shanghai Stock Exchange (SSE) and on the Shenzhen Stock Exchange (SZSE) from 2006 to 2012. We obtain the necessary financial information from the China Securities Markets and Accounting Research Database (CSMAR). We hand-collect all CSR reports from the corporate websites and the *Cninfo* website: 2829 reports in total. Since the mandatory CSR reporting came into force in 2008—and all firms that met the disclosure requirements by the SSE and SZSE must disclose CSR reports from 2008—we code 2006–2007 as the pre-regulation years and 2008–2012 as the post-regulation years.

We eliminate 3331 observations with initial public offerings after 2007. We eliminate 232 observations pertaining to financial services companies. We exclude firms that were not subject to mandatory disclosure *every* year from 2008 to 2012 (1562 firm-year observations). We further eliminate 654 firm-years with missing data. Our control firms are those that are not subject to the CSR regulation. Therefore, we delete 1254 firm-year observations corresponding to firms issuing CSR reports voluntarily after 2008. The final sample comprises 6982 firm-year observations (for 1079 firms). See panel A of Table 1 for details regarding the sample selection screens.

As shown in panel B of Table 1, our treatment group consists of 1585 firm-years representing 236 firms and our control group consists of 5397 firm-years representing 843 firms. We identify a firm as an SOE if the ultimate controlling owner is the government, and as an NSOE otherwise. Our treatment group consists of 183 SOEs and 53 NSOEs.

Panel B of Table 1 presents the distribution of sample firms across industries. The industry composition of our sample is similar to that of the CSMAR

Table 1 Sample Selection

Panel A: Full sample

	Number of firm-years
Total firm-year observations available on CSMAR during 2006–2012	14,015
Less:	
Companies that initially go public after 2007	(3331)
Financial services companies	(232)
Companies that are not subject to regulation every year	(1562)
Observations with missing variables	(654)
Companies that voluntarily disclose CSR after 2008	(1254)
Total firm-year observations in the final sample	6982

Panel B: Industry distribution

	Full		Treatment		Control	
Industry	#	%	#	%	#	%
Agriculture, forestry & fishing	153	2.19	21	1.32	132	2.45
Mining	206	2.95	73	4.61	133	2.46
Manufacturing	3893	55.75	872	55%	3021	55.98
Food and beverage	324	4.64	76	4.79	248	4.60
Textile and apparel	271	3.88	28	1.77	243	4.50
Furniture	31	0.44	0	0	31	0.57
Paper	119	1.7	21	1.32	98	1.82
Chemistry	731	10.47	98	6.18	633	11.73
Electronic	273	3.91	70	4.42	203	3.76
Fabricated metal products	569	8.15	178	11.23	391	7.24
Equipment	1034	14.81	289	18.23	745	13.80
Pharmaceutical	497	7.12	98	6.18	399	7.39
Other	44	0.63	14	0.88	30	0.56
Utilities	388	5.56	120	7.57	268	4.97
Construction	117	1.68	35	2.21	82	1.52
Transportation	292	4.18	143	9.02	149	2.76
Information and technology	309	4.43	70	4.42	239	4.43
Wholesale trade	522	7.48	54	3.41	468	8.67
Real estate	534	7.65	96	6.06	438	8.12
Services	240	3.44	31	1.96	209	3.87
Entertainment	92	1.32	21	1.32	77	1.42
Conglomerates	236	3.38	49	3.09	181	3.35
Total	6982	100	1585	100	5397	100

Note: This table adopts CSRC 23-industry classification.

population. The most heavily represented industry is manufacturing, which comprises 55.75% of sample observations. We note that the number of firm-year observations is likely to vary across different tests depending on the availability of data with respect to the variables involved.

We hand-collect CSR expenditures directly from mandatory stand-alone CSR reports in the post-regulation period. In addition, we manually collect data on CSR expenditures *voluntarily* reported in the notes to financial statements for both the treatment and control groups in the pre-regulation period and for the control group in the post-regulation period.[8] We assume that a firm that does not report any CSR expenditure voluntarily in the notes in the pre-regulation period does not incur any such expenditure (i.e., we set the CSR expenditure for that firm to be zero). This assumption reflects the premise that any firm incurring CSR expenditures has no reason not to disclose—there are no disclosure costs per se.

Our data reveal that a large percentage of treatment firms in our sample *voluntarily* discloses CSR information in the notes to financial statements even before the regulation, which raises an important question as to whether the CSR disclosure mandate should make any difference if CSR information is already being voluntarily disclosed. One possibility is that mandated disclosures provide broader and more detailed information than pre-regulation voluntary disclosures. Moreover, stand-alone disclosure reports likely bring greater and more direct scrutiny. Therefore, firms subject to the regulation could potentially increase CSR expenditures if they believe that their CSR expenditures would be perceived as inadequate relative to acceptable norms (Appendix A presents examples of pre- and post-regulation CSR disclosures by firms that are subject to the mandate).

To the extent treatment and control firms differ systematically pre-regulation, any post-regulation differences may not be attributable to the regulation (Barrot 2015). Clearly, some systematic differences are bound to exist because the regulation mandates disclosure only for firms meeting certain criteria, and thus the treatment and control groups are "non-random."[9] As such, any matching procedure used to identify a suitable control group would be necessarily handicapped to the extent that the source of this non-randomness (i.e., the criteria used to mandate disclosure) is inherently related to the drivers of the real effects of the regulation.[10]

As in Chen et al. (2018), we adopt a difference-in-differences approach along with a set of control variables based on prior literature to account for other systematic differences between treatment and control firms. Specifically, we estimate the following specification (see Appendix B for variable definitions):

$$\begin{aligned} REALEFF_{i,t} = {} & \alpha_0 + \alpha_1 TMT + \alpha_2 POST \\ & + \alpha_3 POST \times TMT + \alpha_4 SOE_{i,t} \\ & + \alpha_5 SIZE_{i,t} + \alpha_6 LOSS_{i,t} + \alpha_7 CFO_{i,t} \\ & + \alpha_8 LEV_{i,t} + \alpha_9 GROWTH_{i,t} \\ & + \alpha_{10} CROSSLIST_{i,t} + \alpha_{11} OWNERSHIP_{i,t} \\ & + \alpha_{12} AGE_{i,t} + FIRM\,FE + \varepsilon_{i,t}. \end{aligned} \tag{1}$$

The dependent variable *REALEFF* represents CSR expenditures (*CSR_EXP*), CSR *capitalized* expenditures (*CSR_CAP)*, cost of goods sold (*COGS*), total operating expenses, and CSR special items (*CSR_NOE*). Because CSR expenditures could involve long-term investments in CSR activities, we deflate *CSR_EXP* and *CSR_CAP* by total assets. Because COGS, total operating expenses, and *CSR_NOE* are income statement items, we deflate them by sales.

TMT is a dummy variable that equals one for treatment firms and zero for control firms. *POST* is a dummy variable that equals one if a firm-year is 2008 or thereafter, and zero otherwise. *SOE* is a dummy variable that equals one if the firm is a state-owned enterprise, and zero otherwise. We include firm size (*SIZE*), loss (*LOSS*), operating cash flow (*CFO*), leverage (*LEV*), growth (*GROWTH*), and firm age (*AGE*) as control variables following prior literature (Biddle and Hilary 2006, Biddle et al. 2009, Chen et al. 2011). Following Shen et al. (2012), we use two proxies for corporate governance—*CROSSLIST* and *OWNERSHIP*. *CROSSLIST* is an indicator variable that equals one if a firm issues shares to foreign investors and zero otherwise. *OWNERSHIP* is the percentage of ownership held by the largest shareholder. We include firm fixed effects (*FIRM FE*) to control for systematic differences between treatment and control firms unrelated to the regulation.

In addition to this difference-in-differences approach, we present a "parallel trends" test by allowing the coefficient of interest to vary over time (Bertrand and Mullainathan 2003). This test allows us to determine whether any evidence of systematic difference in real effects between treatment and control firms can be attributed to pre-regulation differences. Specifically, we estimate the following model:[11]

$$\begin{aligned} REALEFF_{i,t} = {} & \beta_t + \beta_i + \beta_1 TMT_t^T + \beta_{1B} BEFORE \\ & + \beta_{1A} AFTER + \beta_2 SOE_{i,t} + \beta_3 SIZE_{i,t} \\ & + \beta_4 LOSS_{i,t} + \beta_5 CFO_{i,t} + \beta_6 LEV_{i,t} \\ & + \beta_7 GROWTH_{i,t} + \beta_8 CROSSLIST_{i,t} \\ & + \beta_9 OWNERSHIP_{i,t} + \beta_{10} AGE_{i,t} \\ & + FIRM\,FE + \varepsilon_{i,t}. \end{aligned} \tag{2}$$

In this specification, the variable TMT^T assumes a value of one for the treatment firms and zero otherwise. The variable *BEFORE* assumes a value of one for the treatment firms in 2007 and zero otherwise. The variable *AFTER* assumes a value of one for treatment firms in 2008 or thereafter, and zero otherwise. Thus, the coefficient β_1 captures the difference in the *REALEFF* variable between treatment and control firms in 2006, the first year of the pre-regulation period in our sample. The coefficient β_{1B} captures the *incremental* difference (relative to 2006) in the *REALEFF* variable between treatment and control firms before regulation. Similarly, the coefficient β_{1A} captures the *incremental* difference (relative to 2006) in *REALEFF* between treatment and control firms after regulation. With this specification, any significant shifts in the post-regulation period as reflected in the coefficient β_{1A} can be attributed causally to the CSR regulation.

4. Results

Table 2 presents descriptive statistics on key variables. Referring to panel A, NSOE treatment firms are associated with significant increases in CSR expenditures (unscaled *CSR_EXP*) post-regulation. Both the mean and median values of these expenditures show an increase. We do not find a mean effect for NSOE control firms. Panel B reveals that, for SOEs, both the mean and median values of these expenditures show an increase for both treatment and control firms. Observe also that *COGS* (as a percentage of sales) does not show any increase for NSOE treatment firms but does show a statistically significant increase for SOE treatment firms. The mean values of the unscaled non-core expenses (*NOE*) and CSR special items (*CSR_NOE*) are also higher for both NSOE and SOE treatment firms, but not for control firms. However, when scaled by sales (not tabulated), these increases are not statistically significant for SOEs. Price-to-book ratio (*P/B*) decreases for both SOEs and NSOEs from pre-regulation to post-regulation, but this decline is more pronounced for treatment firms. It is clear from these descriptive statistics that the CSR regulation has measurable real effects on the treatment firms.

4.1. Direct CSR Expenditures

We next examine the effects of the regulation on CSR expenditures using the difference-in-differences methodology by estimating Equation (1) using *CSR_EXP* as the dependent variable. If the disclosure mandate stimulates CSR activities, we should find an increase in CSR expenditures for treatment firms relative to control firms in the post-regulation period. We estimate Equation (1) for SOEs and NSOEs and present the results in panel A of Table 3.

Referring to column (3), the coefficient on *TMT* is negative ($\alpha_1 = -0.0024$, $t = -2.63$, $p < 0.01$), indicating that in the pre-regulation period, CSR expenditures are on average *lower* for treatment NSOEs. The coefficient α_3 is positive and significant ($\alpha_3 = 0.0025$, $t = 2.90$, $p < 0.01$), which suggests that CSR expenditure is significantly higher for treatment firms after the mandate than before the mandate as compared to control firms. Column (2) presents results for the SOE subsample. The results are similar with respect to the main coefficient of interest, α_3. On average, an SOE increased its CSR expenditures by ¥78 million (US$11 million) and an NSOE increased its CSR expenditures by ¥19 million (US$2.7 million) post-regulation. Therefore, the mandatory CSR reporting appears to have induced an

Table 2 Descriptive Statistics

Panel A: NSOEs

	TMT			Control		
	Pre- Mean (Median)	Post- Mean (Median)	Diff. *t*-stat. (*z*-stat.)	Pre- Mean (Median)	Post- Mean (Median)	Diff. *t*-stat. (*z*-stat.)
COGS	0.731	0.728	−0.13	0.758	0.738	−2.34**
	(0.776)	(0.766)	(−0.11)	(0.795)	(0.781)	(−1.98)**
P/B	5.451	3.364	−6.09***	5.411	4.495	−3.07***
	(4.872)	(2.802)	(−6.68)***	(4.479)	(3.467)	(−6.21)***
SIZE	21.757	22.558	7.13***	20.760	21.152	8.07***
	(21.727)	(22.434)	(6.02)***	(20.735)	(21.094)	(−7.57)***
ROE	0.165	0.139	−2.01**	0.028	0.053	1.69*
	(0.147)	(0.118)	(−2.80)***	(0.057)	(0.061)	(0.12)
CSR_EXP	8.694	27.575	2.66***	1.057	22.444	1.10
	(0.000)	(2.454)	(8.49)***	(0.005)	(0.108)	(9.06)***
AB_NOI	82.559	213.395	2.31**	−32.532	−42.882	−1.89*
	(−12.803)	(−10.565)	(0.53)	(−30.459)	(−52.416)	(−7.23)***
NOE	7.885	20.557	3.38***	8.956	6.712	−1.39
	(2.189)	(4.820)	(3.78)***	(1.447)	(1.794)	(1.05)
CSR_NOE	0.637	6.070	2.10**	0.260	5.916	1.12
	(0.000)	(0.230)	(8.22)***	(0.000)	(0.050)	(9.20)***

Panel B: SOEs

	TMT			Control		
	Pre- Mean (Median)	Post- Mean (Median)	Diff. *t*-stat. (*z*-stat.)	Pre- Mean (Median)	Post- Mean (Median)	Diff. *t*-stat. (*z*-stat.)
COGS	0.727	0.751	2.45***	0.799	0.792	−1.10
	(0.755)	(0.793)	(2.90)***	(0.830)	(0.826)	(−0.50)
P/B	5.096	2.833	−12.60***	4.654	3.793	−4.86***
	(4.250)	(2.273)	(−15.09)***	(3.840)	(2.928)	(−11.29)***
SIZE	22.438	23.154	9.40***	21.296	21.690	10.99***
	(22.368)	(23.131)	(8.76)***	(21.310)	(21.670)	(10.26)***
ROE	0.163	0.116	−6.40***	0.044	0.035	−0.93
	(0.150)	(0.108)	(−7.18)***	(0.054)	(0.056)	(0.68)
CSR_EXP	21.594	99.438	6.25***	4.240	14.293	3.99***
	(0.000)	(3.468)	(13.08)***	(0.025)	(0.200)	(10.58)***
AB_NOI	57.999	177.910	2.85**	−23.578	−39.636	−2.89***
	(−17.366)	(−22.720)	(−0.41)	(−29.792)	(−46.103)	(−5.56)***
NOE	21.105	36.309	2.62***	7.921	8.770	0.75
	(4.837)	(7.773)	(4.71)***	(1.902)	(2.401)	(3.31)***
CSR_NOE	0.648	9.524	3.87***	0.217	1.506	1.45
	(0.000)	(0.100)	(14.46)***	(0.007)	(0.057)	(8.85)***

Note: ***, **, and * denote two-tailed significance at the 1%, 5%, and 10% level, respectively.

economically significant increase in CSR expenditures for both NSOE and SOEs.

To investigate SOEs further, we partition the SOE subsample into a "High" group and a "Low" group where we use the percentage of state ownership of ultimate controlling shareholder as the partition criterion. An SOE whose ultimate controlling shareholder's ownership is above (below) the median percentage is classified into the "High" ("Low") group. The mean (median) state ownership of the High group is 50.8% (49.8%), whereas that of the Low group is 25.7% (26.1%). As we hypothesize in H2, SOEs that are tightly controlled by the government (the "High" group) may already be incurring significant amount of CSR expenditures in the past and, therefore, may have no incentive to boost its CSR expenditures post-regulation. In contrast, an SOE in which the government has relatively less stake (the "Low" group) may be similar to an NSOE in behavior. We estimate Equation (1) for "High" and "Low" SOE groups separately and present these results in panel B of Table 3.

Referring to Panel B, the difference-in-differences coefficient α_3 is positive and significant for both

Table 3 Effects of CSR Disclosure on CSR Expenditures

Panel A: SOEs vs. NSOEs

Dependent var.		Column (1)		Column (2)		Column (3)	
		FULL		SOE		NSOE	
CSR_EXP		Coeff.	*t* value	Coeff.	*t* value	Coeff.	*t* value
CONSTANT	α_0	0.0012	0.45	0.0008	0.19	0.0061	1.87*
TMT	α_1	−0.0007	−0.60	−0.0001	−0.05	−0.0024	−2.63***
POST	α_2	0.0004	2.47**	0.0006	2.32**	0.0003	1.16
POST × *TMT*	α_3	0.0015	3.74***	0.0013	2.54**	0.0025	2.90***
SOE	α_4	0.0001	0.46				
SIZE	α_5	−0.0001	−0.76	−0.0001	−0.37	−0.0004	−1.97**
LOSS	α_6	−0.0004	−2.57***	−0.0005	−2.94***	−0.0003	−1.35
CFO	α_7	0.0001	0.47	−0.0002	−0.66	0.0002	1.12
LEV	α_8	−0.0003	−0.55	−0.0002	−0.27	−0.0002	−0.16
GROWTH	α_9	0.0000	0.49	0.0001	0.59	0.0001	0.83
CROSSLIST	α_{10}	−0.0004	−0.40	0.0006	0.85	0.0005	0.82
OWNERSHIP	α_{11}	0.0018	1.54	0.0007	0.40	0.0012	1.26
AGE	α_{12}	0.0005	1.12	0.0005	0.64	0.0006	1.00
FIRM FE		Yes		Yes		Yes	
Observations		6982		4562		2420	
Adj. R^2		0.441		0.449		0.437	

Panel B: Impact of state ownership

Dependent var.		Column (1)		Column (2)	
		High		Low	
CSR_EXP		Coeff.	*t* value	Coeff.	*t* value
CONSTANT	α_0	−0.0048	−0.53	0.0158	2.69***
TMT	α_1	0.0406	31.27***	0.0033	1.58
POST	α_2	0.0003	0.82	0.0006	2.11**
POST × *TMT*	α_3	0.0017	2.43**	0.0014	1.89*
SIZE	α_4	0.0002	0.33	−0.0008	−2.58***
LOSS	α_5	−0.0003	−0.97	−0.0006	−2.61****
CFO	α_6	0.0001	0.27	−0.0003	−1.07
LEV	α_7	−0.0003	−0.24	0.0006	0.60
GROWTH	α_8	0.0001	0.36	0.0000	0.02
CROSSLIST	α_9	−0.0001	−0.05	−0.0008	−0.92
OWNERSHIP	α_{10}	−0.0009	−0.24	−0.0038	−0.93
AGE	α_{11}	0.0011	0.90	0.0003	0.35
FIRM FE		Yes		Yes	
Observations		2280		2282	
Adj. R^2		0.482		0.447	

Note: An SOE whose state ownership is above (below) the median percentage is classified into the "High" ("Low") group. The *t*-statistics are adjusted for heteroscedasticity based on White's (1980) approach. ***, **, and * denote two-tailed significance at the 1%, 5%, and 10% level, respectively.

"High" and "Low" SOE groups (for the "High" group, $\alpha_3 = 0.0017$, $t = 2.43$, $p < 0.01$; for the "Low" group, $\alpha_3 = 0.0014$, $t = 1.89$, $p < 0.10$). This finding suggests that compared to the control firms, treatment SOEs with both high and low state ownerships ramp up their CSR expenditures post-regulation.

Turning now to the test of parallel trends in Table 4 corresponding to estimation of Equation (2). Referring to column (3) of Panel A, the coefficient of TMT^T is statistically significant for NSOEs ($\beta_1 = -0.0026$, $t = -2.28$, $p < 0.05$). This negative coefficient implies that treatment NSOEs incur less CSR expenditures than control firms in the base year of 2006. Moreover, there appears to be no incremental difference between treatment and control firms in the year 2007 relative to 2006, as indicated by the coefficient of *BEFORE*. This finding indicates there are no trend differences between NSOE treatment firms and control firms prior to regulation. However, columns (1) and (2) reveal the presence of some *counter* trend in SOE firms. In particular, the negative coefficient of *BEFORE* for the SOE subsample suggests that these firms seem to have *decreased* CSR expenditures in 2007 relative to 2006.

Table 4 Effects of CSR Disclosure on CSR Expenditures: Parallel Trend Tests

Panel A: SOEs vs. NSOEs

Dependent var. *CSR_EXP*		Column (1) FULL Coeff.	Column (1) FULL *t* value	Column (2) SOE Coeff.	Column (2) SOE *t* value	Column (3) NSOE Coeff.	Column (3) NSOE *t* value
CONSTANT	β_0	−0.0011	−0.43	−0.0033	−0.83	0.0057	1.70*
TMT^T	β_1	0.0011	0.98	0.0011	0.77	−0.0026	−2.28**
BEFORE	β_{1B}	−0.0016	−2.38**	−0.0020	−2.64***	0.0000	0.03
AFTER	β_{1A}	0.0009	1.50	0.0004	0.59	0.0026	2.34**
SOE	β_2	0.0001	0.45				
SIZE	β_3	−0.0001	−0.67	−0.0001	−0.20	−0.0004	−2.09**
LOSS	β_4	−0.0004	−2.39**	−0.0005	−2.73***	−0.0003	−1.27
CFO	β_5	0.0000	0.25	−0.0002	−0.86	0.0002	1.11
LEV	β_6	−0.0004	−0.69	−0.0003	−0.41	−0.0002	−0.21
GROWTH	β_7	0.0000	0.45	0.0001	0.59	0.0001	0.76
CROSSLIST	β_8	−0.0001	−0.06	0.0014	2.56***	0.0005	0.72
OWNERSHIP	β_9	0.0018	1.56	0.0007	0.39	0.0013	1.32
AGE	β_{10}	0.0012	3.90***	0.0014	2.81***	0.0010	2.85***
FIRM FE		Yes		Yes		Yes	
Observations		6982		4562		2420	
Adj. R^2		0.442		0.451		0.436	

Panel B: impact of state ownership

Dependent var. *CSR_EXP*		Column (1) High Coeff.	Column (1) High *t* value	Column (2) Low Coeff.	Column (2) Low *t* value
CONSTANT	β_0	−0.0108	−1.13	0.0100	2.01**
TMT^T	β_1	0.0434	28.91***	0.0048	2.56***
BEFORE	β_{1B}	−0.0030	−2.94***	−0.0000	−0.00
AFTER	β_{1A}	0.0002	0.18	0.0017	1.87*
SIZE	β_3	0.0003	0.53	−0.0008	−2.72***
LOSS	β_4	−0.0003	−0.97	−0.0005	−2.36**
CFO	β_5	0.0000	0.04	−0.0003	−1.05
LEV	β_6	−0.0004	−0.31	0.0005	0.49
GROWTH	β_7	0.0001	0.26	−0.0000	−0.02
CROSSLIST	β_8	0.0023	1.43	0.0032	2.37**
OWNERSHIP	β_9	−0.0013	−0.34	−0.0038	−0.92
AGE	β_{10}	0.0018	1.99**	0.0013	2.52**
FIRM FE		Yes		Yes	
Observations		2280		2282	
Adj. R^2		0.486		0.446	

Note: An SOE whose state ownership is above (below) the median percentage is classified into the "High" ("Low") group. The *t*-statistics are adjusted for heteroscedasticity based on White's (1980) approach. ***, **, and * denote two-tailed significance at the 1%, 5%, and 10% level, respectively.

Our results support H1: as can be seen from column (3), the coefficient of *AFTER* is significantly positive for the NSOEs ($\beta_{1A} = 0.0026$, $t = 2.34$, $p < 0.05$). That is, NSOEs subject to the regulation increase CSR expenditures following the regulation.

Panel B of Table 4 presents the results for SOEs with high and low state ownership. As this panel indicates, only SOEs with low state ownership appear to have increased their CSR expenditures post-regulation—the coefficient of *AFTER* is positive and significant ($\beta_{1A} = 0.0017$, $t = 1.87$, $p < 0.10$). Moreover, we do not find such an increase for SOEs with high state ownership when we control pre-regulation trend differences. Taken together, these results offer support to H2 that SOE firms with low state ownership appear to increase CSR expenditures in the post-regulation period.

4.2. Core Operating Costs

Turning now to core operations, in Table 5, we report the results from estimating Equation (1) using cost of goods sold (*COGS*) and total operating expenses as our dependent variables (H3A).

Referring to panel A, the difference-in-differences effect on *COGS* is significantly positive for the full sample ($\alpha_3 = 0.0288$, $t = 6.27$, $p < 0.01$). However, referring to columns (2) and (3), it appears that the results for the full sample are attributable to the SOE subsample. Specifically, the difference-in-differences coefficient is significantly positive for the SOE subsample ($\alpha_3 = 0.0268$, $t = 5.10$, $p < 0.01$), but not so for NSOEs. Thus, the CSR regulation seems to have resulted in higher *COGS* for SOEs post-regulation, but not for NSOEs. Moreover, panel B reveals that among SOEs, *COGS* increases only for those with high state ownership, and not for those with low state ownership. In panels C and D, we replicate the results of panels A and B using *total operating expenses* as the dependent variable in place of *COGS* (we compute total operating expenses as the sum of *cost of goods sold* and *selling, general, and administrative expenses*). The results are strikingly similar to the results reported for *COGS* in panels A and B.

We next perform the parallel trends test for *COGS* and total operating expenses by estimating Equation (2). The findings (untabulated) from this test are in line with our inferences from Table 5. The result that the CSR regulation has precipitated higher production costs for SOEs with high state ownership is in line with the results in Chen et al. (2018), Brammer and Millington (2008), and Hazilla and Kopp (1990). It supports the narrative that prior to regulation, government-controlled SOEs could be requiring labor to subsidize production to be extremely competitive in the global market. Faced with increasing global scrutiny of the plight of the labor force, the government is perhaps using the CSR regulation as a credible "self-commitment" mechanism to have SOEs increase wages, offer additional employee benefits, and raise the standard of living of the workforce (Barnett and Salomon 2012).

4.3. Non-core CSR Items

The results reported in Tables 3–5 indicate that NSOEs subject to the regulation increase their total CSR expenditures, but not their core operating costs (unlike SOEs). One possibility is that their CSR expenditures are non-core in nature—such as donations to charities and community projects. NSOEs may be more inclined to engage in such activities so as to lessen their perceived negative wealth-transfer effect on shareholder value. Indeed, evidence suggests that non-core expenses and special items are not considered as value relevant as core operating items (McVay 2006).

To investigate this possibility further, we manually identify CSR special items (*CSR_NOE*) from the financial statements for our treatment and control firms. Referring to panel A of Table 2, these expenses (unscaled) increase from a mean (median) of ¥0.637 million (¥0.000 million) in the pre-regulation period to ¥6.070 million (¥0.230 million) for NSOEs subject to the regulation. Total non-core expenses (*NOE*) also increase for these firms. Interestingly, however, the mean of the industry-adjusted non-core income (*AB_NOI*) also increase post-regulation, indicating that increases in non-core revenues and gains more than overwhelm increases in CSR expenditures. To shed additional insight, we estimate Equation (1) using industry-adjusted non-core income (*AB_NOI*), total non-core expenses (*NOE*), and CSR special items (*CSR_NOE*) as the dependent variables. Table 6 presents these results.

As panel A indicates, the difference-in-differences coefficient is significantly positive in all three columns for the full sample. That is, industry-adjusted non-core income (*AB_NOI*), total non-core expenses (*NOE*), and, in particular, CSR special items (*CSR_NOE*) increase significantly for treatment firms relative to control firms, post-regulation. More importantly, referring to column (3) of panels B–D, the coefficient α_3 is positive and significant for NSOEs and SOEs with low state ownership but not for SOEs with high state ownership, supporting H3B.[12] Moreover, the result that industry-adjusted non-core income (*AB_NOI*) has increased significantly for treatment NSOEs post-regulation (panel B of Table 6) supports the narrative that these firms could be strategically recording increases in non-core revenues and gains to offset increases in non-core CSR expenses, thereby lessening their negative impact on perceived shareholder value.

Firms may also capitalize some portion of the CSR expenditures to delay the impact on the income statement. A casual inspection of CSR disclosures reveals that some CSR expenditures are immediately expensed while other investment-type expenditures are capitalized. We re-estimate Equations (1) and (2) using the capitalized portion of CSR expenditure (*CSR_CAP*) as the dependent variable. We hand-collect this information from the CSR reports for treatment firms post-regulation and from the notes to financial statements pre-regulation. Panels E–G indicate once again that these expenditures show a significant increase only for NSOEs after controlling for pre-regulation trends, thereby supporting H3C, although we do not find a similar result for SOEs (with high or low state ownership).

Taken together with the results of Table 5, the findings speak to the *types* of CSR initiatives that the disclosure mandate has evoked in NSOEs and SOEs with *low* state ownership. Specifically, the CSR initiatives of these firms do not appear to relate to core operating activities but are more non-core in nature. Moreover, the increases in non-core revenues and gains (indicated by the positive and significant coefficient α_3 in

Table 5 Effects of CSR Disclosure on Core Operating Costs

Panel A: Effects of CSR disclosure on cost of goods sold

Dependent var. *COGS*		Column (1) FULL Coeff.	Column (1) FULL *t* value	Column (2) SOE Coeff.	Column (2) SOE *t* value	Column (3) NSOE Coeff.	Column (3) NSOE *t* value
CONSTANT	α_0	0.3240	5.45***	1.5098	20.09***	1.2902	8.87***
TMT	α_1	0.2724	7.95***	−0.4894	−16.66***	−0.2075	−5.14***
POST	α_2	−0.0140	−4.06***	−0.0114	−3.14***	−0.0142	−1.99**
POST × *TMT*	α_3	0.0288	6.27***	0.0268	5.10***	0.0102	1.21
SOE	α_4	0.0326	3.60***				
SIZE	α_5	−0.0080	−2.03**	−0.0086	−1.86*	−0.0117	−1.70*
LOSS	α_6	0.0684	18.91***	0.0657	16.63***	0.0689	9.87***
CFO	α_7	−0.0029	−0.36	−0.0132	−1.42	0.0048	0.37
LEV	α_8	0.0797	5.07***	0.0853	4.36***	0.0614	2.40**
GROWTH	α_9	0.0046	1.06	0.0085	1.71*	0.0018	0.26
CROSSLIST	α_{10}	−0.1710	−6.99***	−0.5821	−51.15***	−0.0958	−2.42**
OWNERSHIP	α_{11}	−0.0909	−3.68***	−0.0446	−1.36	−0.1490	−3.34***
AGE	α_{12}	0.0141	1.85*	0.0309	3.67***	−0.0064	−0.46
FIRM FE		Yes		Yes		Yes	
Observations		6982		4562		2420	
Adj. R^2		0.798		0.840		0.763	

Panel B: Effects of CSR disclosure on cost of goods sold for SOEs: Impact of state ownership

Dependent var. *COGS*		Column (1) High Coeff.	Column (1) High *t* value	Column (2) Low Coeff.	Column (2) Low *t* value
CONSTANT	α_0	1.1116	8.50***	0.9507	7.08***
TMT	α_1	0.0037	0.26	−0.6859	−15.64***
POST	α_2	−0.0164	−2.91***	−0.0002	−0.04
POST × *TMT*	α_3	0.0352	5.20***	0.0101	1.13
SIZE	α_4	−0.0175	−2.78***	0.0209	2.96***
LOSS	α_5	0.0735	11.29***	0.0534	11.59***
CFO	α_6	−0.0061	−0.46	−0.0207	−1.55
LEV	α_7	0.0882	3.37***	0.0847	3.44***
GROWTH	α_8	0.0016	0.26	0.0171	2.50**
CROSSLIST	α_9	−0.0775	−3.29***	−0.4386	−28.40***
OWNERSHIP	α_{10}	0.0194	0.35	0.0467	0.71
AGE	α_{11}	0.0623	5.34***	−0.0279	−2.10**
FIRM FE		Yes		Yes	
Observations		2280		2282	
Adj. R^2		0.856		0.860	

Panel C: Effects of CSR disclosure on total operating expenses

Dependent var. Total operating expenses		Column (1) FULL Coeff.	Column (1) FULL *t* value	Column (2) SOE Coeff.	Column (2) SOE *t* value	Column (3) NSOE Coeff.	Column (3) NSOE *t* value
CONSTANT	α_0	2.1354	24.21***	1.8726	21.11***	1.8273	8.48***
TMT	α_1	−0.3795	−8.89***	−0.3205	−9.63***	−0.0525	−1.35
POST	α_2	−0.0118	−2.45**	−0.0074	−1.56	−0.0060	−0.60
POST × *TMT*	α_3	0.0273	5.32***	0.0242	4.69***	0.0047	0.43
SOE	α_4	0.0154	1.05				
SIZE	α_5	−0.0455	−7.37***	−0.0274	−4.92***	−0.0478	−4.32***
LOSS	α_6	0.1185	21.77***	0.0968	17.76***	0.1366	12.56***
CFO	α_7	−0.0635	−4.74***	−0.0553	−3.75***	−0.0656	−3.08***

(continued)

Table 5 ***(continued)***

Panel C: Effects of CSR disclosure on total operating expenses

		Column (1)		Column (2)		Column (3)	
		FULL		SOE		NSOE	
Dependent var. Total operating expenses		**Coeff.**	***t* value**	**Coeff.**	***t* value**	**Coeff.**	***t* value**
LEV	α_8	0.1036	4.12***	0.1223	4.75***	0.0367	0.77
GROWTH	α_9	−0.0554	−9.15***	−0.0438	−6.77***	−0.0621	−6.09***
CROSSLIST	α_{10}	−0.2716	−17.43***	−0.4648	−30.94***	0.0755	1.99**
OWNERSHIP	α_{11}	−0.1566	−4.44***	−0.1056	−2.90***	−0.1175	−1.83*
AGE	α_{12}	0.0392	3.49***	0.0219	2.09**	0.0312	1.44
FIRM FE		Yes		Yes		Yes	
Observations		6982		4562		2420	
Adj. R^2		0.675		0.752		0.656	

Panel D: Effects of CSR disclosure on total operating expenses for SOEs: Impact of state ownership

		Column (1)		Column (2)	
		High		Low	
Dependent var. Total operating expenses		**Coeff.**	***t* value**	**Coeff.**	***t* value**
CONSTANT	α_0	1.6927	11.66***	1.5025	7.86***
TMT	α_1	−0.0163	−0.98	−0.4650	−7.44***
POST	α_2	−0.0151	−2.41**	0.0050	0.72
POST × *TMT*	α_3	0.0354	5.33***	0.0082	0.99
SIZE	α_4	−0.0408	−5.78***	−0.0065	−0.64
LOSS	α_5	0.0946	12.29***	0.0858	11.13***
CFO	α_6	−0.0393	−2.17**	−0.0643	−2.66***
LEV	α_7	0.1031	3.28***	0.1381	3.04***
GROWTH	α_8	−0.0287	−3.77***	−0.0576	−5.08***
CROSSLIST	α_9	−0.0438	−1.58	−0.3539	−16.89***
OWNERSHIP	α_{10}	−0.0004	−0.01	−0.0727	−0.70
AGE	α_{11}	0.0664	4.67***	−0.0331	−1.82*
FIRM FE		Yes		Yes	
Observations		2280		2282	
Adj. R^2		0.794		0.739	

Note: An SOE whose state ownership is above (below) the median percentage is classified into the "High" ("Low") group. The *t*-statistics are adjusted for heteroscedasticity based on White's (1980) approach. ***, **, and * denote two-tailed significance at the 1%, 5%, and 10% level, respectively.

column 1 of panel B) likely lessen the negative valuation impact of the increases in non-core CSR expenditures.

4.4. Other Discretionary Expenditures

Given that firms subject to the CSR regulation appear to increase their CSR expenditures, a question that arises is whether they scale back their other discretionary expenditures such as R&D, advertising, and SG&A expenses in order to devote more resources to CSR activities. To address this issue, we estimate normal discretionary expenditure levels in these items using the following model adapted from Roychowdhury (2006):

$$DISEXP_{i,t} = \theta_0 + \theta_1 \frac{1}{ASSETS_{i,t-1}} + \theta_2 \frac{SALES_{i,t-1}}{ASSETS_{i,t-1}} + \varepsilon_{i,t}. \tag{3}$$

In this equation, the dependent variable *DISEXP* is the sum of R&D, advertising, and SG&A expenses. We estimate this equation cross-sectionally for each year from 2006 to 2012 and calculate abnormal discretionary expenditures *AB_DISEXP* as the actual discretional expenditures less the predicted amount from the equation. We then estimate the difference-in-differences specification in Equation (1) using *AB_DISEXP* as the dependent variable (results untabulated).

We find that the difference-in-differences coefficient is negative and significant for the full sample and for the SOE subsample, but not for the NSOE subsample. Thus, SOEs appears to cut back on their discretionary expenditures post-regulation, but not NSOEs. In other words, faced with increased operating costs following the regulation, SOEs seem to be redirecting funds from discretionary items to CSR initiatives. Interestingly, this result is primarily driven

Table 6 Effects of CSR Disclosure on Non-Core Costs

Panel A: Full sample

		Column (1)		Column (2)		Column (3)	
		AB_NOI		*NOE*		*CSR_NOE*	
Dependent var.		Coeff.	*t* value	Coeff.	*t* value	Coeff.	*t* value
CONSTANT	α_0	8.6660	8.01***	0.1322	4.72***	0.0031	5.36***
TMT	α_1	−1.5155	−2.69***	0.0745	5.28***	−0.0012	−4.70***
POST	α_2	−0.4268	−5.25***	−0.0021	−1.37	0.0004	12.48***
POST × *TMT*	α_3	0.2189	1.77*	0.0055	4.27***	0.0003	5.92***
SOE	α_4	−0.0730	−0.48	−0.0026	−0.56	−0.0000	−0.55
SIZE	α_5	−0.3710	−4.90***	−0.0097	−4.87***	−0.0001	−1.75*
LOSS	α_6	−0.0995	−1.45	0.0106	6.41***	−0.0000	−0.39
CFO	α_7	0.1561	1.20	−0.0051	−1.25	−0.0001	−0.98
LEV	α_8	0.5479	2.03**	0.0351	4.17***	−0.0001	−0.82
GROWTH	α_9	0.1228	1.78*	−0.0061	−4.08***	−0.0001	−3.07***
CROSSLIST	α_{10}	−0.1564	−0.07	0.1850	2.93***	0.0006	1.50
OWNERSHIP	α_{11}	−0.8874	−1.86*	−0.0073	−0.73	0.0001	0.26
AGE	α_{12}	0.9183	5.66***	0.0029	0.87	−0.0007	−8.51***
FIRM FE		Yes		Yes		Yes	
Observations		6982		6982		6982	
Adj. R^2		0.121		0.313		0.253	

Panel B: NSOE

		Column (1)		Column (2)		Column (3)	
		AB_NOI		*NOE*		*CSR_NOE*	
Dependent var.		Coeff.	*t* value	Coeff.	*t* value	Coeff.	*t* value
CONSTANT	α_0	6.7569	2.46**	0.1100	1.58	0.0024	1.72*
TMT	α_1	−1.3660	−3.05***	0.0323	3.03***	0.0001	0.68
POST	α_2	−0.5896	−3.64***	0.0007	0.20	0.0005	7.23***
POST × *TMT*	α_3	0.3737	1.68*	0.0042	1.63	0.0002	2.41**
SIZE	α_4	−0.3190	−2.45**	−0.0076	−2.16**		
LOSS	α_5	−0.3186	−2.26**	0.0155	4.40***	−0.0001	−0.97
CFO	α_6	−0.0242	−0.13	−0.0105	−1.63	0.0000	0.57
LEV	α_7	0.4781	0.97	0.0439	2.65***	−0.0001	−0.97
GROWTH	α_8	0.2418	2.27**	−0.0069	−2.97***	−0.0002	−0.81
CROSSLIST	α_9	−0.7529	−2.12**	0.0394	3.82***	−0.0001	−1.59
OWNERSHIP	α_{10}	−1.3629	−1.62	0.0082	0.43	0.0005	3.00***
AGE	α_{11}	0.9901	3.61***	−0.0042	−0.71	0.0006	1.57
FIRM FE		Yes		Yes		Yes	
Observations		2420		2420		2420	
Adj. R^2		0.122		0.390		0.252	

Panel C: Low-state ownership SOE

		Column (1)		Column (2)		Column (3)	
		AB_NOI		*NOE*		*CSR_NOE*	
Dependent var.		Coeff.	*t* value	Coeff.	*t* value	Coeff.	*t* value
CONSTANT	α_0	8.6505	2.77***	0.0889	1.57	0.0050	3.31***
TMT	α_1	3.4815	3.13***	0.0484	2.01**	−0.0008	−1.54
POST	α_2	−0.2527	−1.91*	−0.0005	−0.21	0.0004	5.13***
POST × *TMT*	α_3	−0.0778	−0.27	0.0010	0.50	0.0005	2.79***
SIZE	α_4	−0.5344	−3.03***	−0.0065	−2.10**	−0.0001	−0.74
LOSS	α_5	−0.0846	−0.80	0.0073	2.79***	−0.0001	−1.34
CFO	α_6	−0.2376	−0.99	0.0011	0.11	0.0001	0.43
LEV	α_7	0.2921	0.47	0.0270	1.60	−0.0001	−0.28

(continued)

Table 6 ***(continued)***

Panel C: Low-state ownership SOE

		Column (1)		Column (2)		Column (3)	
		AB_NOI		*NOE*		*CSR_NOE*	
Dependent var.		**Coeff.**	***t* value**	**Coeff.**	***t* value**	**Coeff.**	***t* value**
GROWTH	α_8	0.0669	0.48	−0.0051	−1.50	−0.0002	−2.87***
CROSSLIST	α_9	−0.1042	−0.31	−0.0037	−0.52	−0.0009	−4.75***
OWNERSHIP	α_{10}	−2.5899	−1.23	0.0106	0.20	−0.0029	−3.21***
AGE	α_{11}	0.7264	1.96**	0.0045	0.68	−0.0007	−3.05***
FIRM FE		Yes		Yes		Yes	
Observations		2282		2282		2282	
Adj. R^2		0.259		0.437		0.207	

Panel D: High-state ownership SOE

		Column (1)		Column (2)		Column (3)	
		AB_NOI		*NOE*		*CSR_NOE*	
Dependent var.		**Coeff.**	***t* value**	**Coeff.**	***t* value**	**Coeff.**	***t* value**
CONSTANT	α_0	7.6822	2.69***	−0.0060	−0.27	−0.0027	−0.18
TMT	α_1	−0.7547	−2.55***	0.0022	0.92	0.0042	1.07
POST	α_2	−0.2220	−1.53	−0.0013	−1.21	0.0022	1.03
POST × *TMT*	α_3	0.1310	0.80	0.0027	2.30**	−0.0004	−0.22
SIZE	α_4	−0.4328	−2.99***	0.0004	0.42	0.0001	0.14
LOSS	α_5	0.0761	0.71	0.0050	3.05***	−0.0005	−1.12
CFO	α_6	0.3807	1.56	−0.0031	−1.53	−0.0007	−0.98
LEV	α_7	1.1744	3.00***	0.0048	1.08	0.0032	0.75
GROWTH	α_8	0.0446	0.36	−0.0000	−0.04	0.0009	0.96
CROSSLIST	α_9	−2.0715	−3.39***	−0.0018	−0.42	0.0041	0.96
OWNERSHIP	α_{10}	1.0013	1.36	0.0056	0.71	−0.0035	−0.88
AGE	α_{11}	0.7573	3.07***	−0.0045	−2.12**	−0.0014	−0.87
FIRM FE		Yes		Yes		Yes	
Observations		2280		2280		2280	
Adj. R^2		0.117		0.346		0.145	

Panel E: Effects of CSR disclosure on expenditure capitalization

		Column (1)		Column (2)		Column (3)	
Dependent var.		*FULL*		*SOE*		*NSOE*	
CSR_CAP		**Coeff.**	***t* value**	**Coeff.**	***t* value**	**Coeff.**	***t* value**
CONSTANT	α_0	0.0034	2.20**	0.0067	2.48**	−0.0003	−0.11
TMT	α_1	−0.0001	−0.11	−0.0002	−0.14	−0.0013	−1.82*
POST	α_2	0.0002	1.45	0.0003	1.94*	−0.0000	−0.24
POST × *TMT*	α_3	0.0011	3.57***	0.0010	2.61***	0.0015	2.61***
SOE	α_4	−0.0001	−0.44				
SIZE	α_5	−0.0001	−1.17***	−0.0003	−1.56	0.0000	0.02
LOSS	α_6	−0.0003	−2.23**	−0.0003	−2.14**	−0.0002	−1.07
CFO	α_7	0.0001	1.21	−0.0000	−0.01	0.0001	1.61
LEV	α_8	−0.0001	−0.29	0.0001	0.08	0.0001	0.15
GROWTH	α_9	0.0001	1.56	0.0001	1.01	0.0001	1.13
CROSSLIST	α_{10}	0.0001	0.14	−0.0002	−0.36	0.0001	0.31
OWNERSHIP	α_{11}	0.0006	0.85	0.0007	0.58	0.0003	0.39
AGE	α_{12}	−0.0004	−1.05	−0.0005	−0.94	0.0000	0.10
FIRM FE		Yes		Yes		Yes	
Observations		6982		4562		2420	
Adj. R^2		0.375		0.359		0.428	

(continued)

Table 6 *(continued)*

Panel F: Parallel trend tests of expenditure capitalization

Dependent var.		***FULL***		***SOE***		***NSOE***	
CSR_CAP		**Coeff.**	***t* value**	**Coeff.**	***t* value**	**Coeff.**	***t* value**
CONSTANT	β_0	0.0024	1.64	0.0044	1.73*	−0.0002	−0.07
TMT^T	β_1	0.0008	1.01	0.0005	0.50	−0.0013	−1.58
BEFORE	β_{1B}	−0.0009	−1.80*	−0.0011	−1.96**	0.0001	0.12
AFTER	β_{1A}	0.0007	1.57	0.0005	0.93	0.0015	2.10**
Control variables		Yes		Yes		Yes	
FIRM FE		Yes		Yes		Yes	
Observations		6982		4562		2420	
Adj. R^2		0.376		0.369		0.428	

Panel G: Effects of CSR disclosure on expenditure capitalization: The impact of state ownership

					Parallel trend tests			
Dependent var.	**High**		**Low**		**High**		**Low**	
CSR_CAP	**Coeff.**	***t* value**	**Coeff.**	***t* value**	**Coeff.**	***t* value**	**Coeff.**	***t* value**
CONSTANT	0.0044	0.73	0.0106	2.36**	0.0034	0.56	0.0071	1.87*
TMT	0.0287	36.00***	0.0017	1.11				
POST	0.0001	0.51	0.0003	1.42				
POST × *TMT*	0.0014	2.85***	0.0008	1.41				
TMT^T					0.0281	31.49***	0.0024	1.77*
BEFORE					−0.0017	−2.24**	0.0001	0.10
AFTER					0.0005	0.73	0.0010	1.40
SIZE	−0.0002	−0.59	−0.0005	−2.16**	−0.0001	−0.42	−0.0005	−2.27**
LOSS	−0.0002	−0.64	−0.0004	−2.23**	−0.0002	−0.67	−0.0004	−2.07**
CFO	0.0002	0.85	−0.0000	−0.15	0.0002	0.69	−0.0000	−0.12
LEV	0.0005	0.46	0.0002	0.27	0.0005	0.44	0.0001	0.20
GROWTH	0.0001	0.57	0.0001	0.49	0.0001	0.49	0.0001	0.46
CROSSLIST	0.0004	0.32	−0.0008	−1.20	−0.0004	−0.40	0.0019	1.87
OWNERSHIP	−0.0002	−0.08	−0.0016	−0.53	−0.0004	−0.18	−0.0016	−0.52
AGE	−0.0004	−0.53	0.0000	0.07	−0.0001	−0.16	0.0006	1.37
FIRM FE	Yes		Yes		Yes		Yes	
Observations	2280		2282		2280		2282	
Adj. R^2	0.366		0.412		0.370		0.411	

Note: An SOE whose state ownership is above (below) the median percentage is classified into the "High" ("Low") group. The *t*-statistics are adjusted for heteroscedasticity based on White's (1980) approach. ***, **, and * denote two-tailed significance at the 1%, 5%, and 10% level, respectively.

by SOEs with low state ownership. One reason could be that SOEs with high state ownership have access to adequate funds compared to SOEs with low state ownership. Moreover, NSOEs do not appear to be engaging in such reallocation of funds, perhaps because reducing discretionary expenditures from their normal levels may be viewed as being value destroying to shareholders.

4.5. Profitability and Valuation

To examine the impact of the CSR regulation on profitability of treatment firms (H4A), we modify the difference-in-differences regression specification of Equation (1) using ROA (net income divided by beginning book value of assets) and ROE (net income divided by beginning book value of equity) as the dependent variable:

$$\begin{aligned} ROA_{(i,t)} \text{ or } ROE_{(i,t)} &= \alpha_0 + \alpha_1 TMT + \alpha_2 POST1 \\ &+ \alpha_3 POST2 + \alpha_4 POST1 \times TMT \\ &+ \alpha_5 POST2 \times TMT + \alpha_6 SOE_{i,t} \\ &+ \alpha_7 SIZE_{i,t} + \alpha_8 CFO_{i,t} \\ &+ \alpha_9 LEV_{i,t} + \alpha_{10} GROWTH_{i,t} \\ &+ \alpha_{11} CROSSLIST_{i,t} + \alpha_{12} OWNERSHIP_{i,t} \\ &+ \alpha_{13} AGE_{i,t} + FIRM\,FE + \varepsilon_{i,t} \end{aligned} \tag{4}$$

In this specification, the dummy variable *POST*1 (*POST*2) equals one for years 2008 and 2009 (years 2010, 2011, and 2012) and zero otherwise. Thus, the coefficient α_4 captures the impact of the regulation on profitability immediately following its implementation, whereas α_5 captures the impact in the longer term. Table 7 presents the results.

Table 7 Effects of CSR Disclosure on Profitability

Panel A: Effects on ROA

Dependent var. *ROA*		Column (1) FULL Coeff.	Column (1) FULL *t* value	Column (2) SOE Coeff.	Column (2) SOE *t* value	Column (3) NSOE Coeff.	Column (3) NSOE *t* value
CONSTANT	α_0	0.013	0.18	–0.399	–4.59***	–0.781	–5.05***
TMT	α_1	–0.674	–23.42***	–0.154	–5.71***	0.126	3.59***
POST1	α_2	–0.010	–3.21***	–0.012	–3.52***	–0.008	–1.36
POST2	α_3	–0.006	–1.30	–0.007	–1.50	–0.004	–0.47
POST1 × *TMT*	α_4	–0.007	–1.74*	–0.006	–1.24	0.003	0.28
POST2 × *TMT*	α_5	–0.037	–9.00***	–0.034	–7.51***	–0.024	–2.60***
SOE	α_6	–0.025	–2.96***				
SIZE	α_7	0.036	8.21***	0.033	6.67***	0.033	4.66***
CFO	α_8	0.004	0.73	–0.002	–0.20	0.008	1.02
LEV	α_9	–0.218	–13.02***	–0.225	–12.19***	–0.203	–6.41***
GROWTH	α_{10}	0.038	10.69***	0.041	8.73***	0.033	6.22***
CROSSLIST	α_{11}	–0.659	–6.62***	–0.041	–3.77***	0.079	2.38**
OWNERSHIP	α_{12}	0.122	4.67***	0.097	3.06***	0.154	3.29***
AGE	α_{13}	–0.017	–2.00**	–0.022	–2.56***	–0.016	–1.04
FIRM FE		Yes		Yes		Yes	
Observations		6982		4562		2420	
Adj. R^2		0.421		0.502		0.394	

Panel B: Effects on ROE

Dependent var. *ROE*		Column (1) FULL Coeff.	Column (1) FULL *t* value	Column (2) SOE Coeff.	Column (2) SOE *t* value	Column (3) NSOE Coeff.	Column (3) NSOE *t* value
CONSTANT	α_0	–1.059	–4.79***	–1.352	–3.98***	–1.754	–3.90***
TMT	α_1	–0.111	–1.20	0.054	1.11	0.450	5.21***
POST1	α_2	–0.022	–2.02**	–0.029	–2.45***	–0.022	–1.05
POST2	α_3	–0.021	–1.47	–0.028	–1.81*	–0.014	–0.48
POST1 × *TMT*	α_4	–0.021	–1.75*	–0.014	–1.07	–0.004	–0.17
POST2 × *TMT*	α_5	–0.065	–5.90***	–0.062	–5.15***	–0.050	–2.13**
SOE	α_6	–0.010	–0.37				
SIZE	α_7	0.056	4.02***	0.067	3.91***	0.056	2.50**
CFO	α_8	–0.002	–0.10	0.019	0.77	–0.023	–0.89
LEV	α_9	–0.200	–3.44***	–0.290	–3.96***	–0.053	–0.56
GROWTH	α_{10}	0.083	7.10***	0.123	7.79***	0.043	2.65***
CROSSLIST	α_{11}	–0.007	–0.20	–0.074	–2.05**	0.262	3.07***
OWNERSHIP	α_{12}	0.265	3.42***	0.119	1.31	0.252	1.89*
AGE	α_{13}	0.008	0.31	–0.004	–0.16	–0.006	–0.13
FIRM FE		Yes		Yes		Yes	
Observations		6982		4562		2420	
Adj. R^2		0.222		0.281		0.238	

Panel C: The impact of state ownership

Dependent var.		ROA High Coeff.	ROA High *t* value	ROA Low Coeff.	ROA Low *t* value	ROE High Coeff.	ROE High *t* value	ROE Low Coeff.	ROE Low *t* value
CONSTANT	α_0	–0.594	–3.86***	–0.191	–1.29	–0.201	–0.55	–0.089	–0.14
TMT	α_1	–0.081	–3.57***	–0.107	–2.29**	–1.441	–12.99***	0.181	0.84
POST1	α_2	–0.009	–1.80*	–0.018	–3.88***	–0.011	–0.65	–0.054	–3.21***
POST2	α_3	–0.010	–1.36	–0.012	–1.92*	–0.012	–0.54	–0.054	–2.46**
POST1 × *TMT*	α_4	–0.007	–1.04	–0.008	–1.25	–0.022	–1.37	–0.002	–0.12

(continued)

Table 7 ***(continued)***

Panel C: The impact of state ownership

		ROA				ROE			
		High		Low		High		Low	
Dependent var.		Coeff.	*t* value	Coeff.	*t* value	Coeff.	*t* value	Coeff.	*t* value
POST2 × *TMT*	α_5	−0.032	−4.93***	−0.029	−4.17***	−0.060	−3.74***	−0.047	−2.14**
SIZE	α_7	0.031	4.36***	0.015	2.09**	0.082	4.06***	0.000	0.01
CFO	α_8	0.002	0.19	−0.008	−0.68	0.080	2.16**	−0.020	−0.65
LEV	α_9	−0.176	−7.18***	−0.286	−10.01***	−0.210	−2.18**	−0.350	−2.97***
GROWTH	α_{10}	0.049	7.02***	0.029	5.18***	0.110	4.76***	0.123	5.81***
CROSSLIST	α_{11}	−0.052	−2.30**	0.037	2.22**	−0.065	−2.08**	0.063	1.04
OWNERSHIP	α_{12}	0.135	1.98**	0.193	2.64***	0.249	1.33	−0.348	−1.13
AGE		−0.028	−2.32**	0.012	0.94	−0.080	−2.22**	0.121	2.66***
FIRM FE		Yes		Yes		Yes		Yes	
Observations		2280		2282		2280		2282	
Adj. R^2		0.547		0.492		0.387		0.250	

Note: *POST1* = 1 if year is 2008 or 2009, otherwise *POST1* = 0; *POST2* = 1 if year is 2010, 2011, or 2012, otherwise *POST2* = 0. An SOE whose state ownership is above (below) the median percentage is classified into the "High" ("Low") group. The *t*-statistics are adjusted for heteroscedasticity based on White's (1980) approach. ***, **, and * denote two-tailed significance at the 1%, 5%, and 10% level, respectively.

In panel A (B), we present the results corresponding to ROA (ROE). We cannot reject the null that the coefficient α_4 is zero in both panels for SOEs and NSOEs. Thus, we do not detect any shift in profitability immediately following the regulation for SOEs and NSOEs (i.e., in years 2008 and 2009). Interestingly, the coefficient α_5 is negative and significant in both panels for both SOEs and NSOEs. That is, profitability declines for SOEs and NSOEs but with a lag indicating that it takes time for any reallocation of resources (to CSR) to affect the bottom line. As noted earlier, Chen et al. (2018) do not observe any profitability effect for NSOEs, because their sample period is shorter than ours, and thus they do not parse out longer-term effects of the regulation from its immediate effects. Moreover, referring to panel C, the coefficient α_5 is negative and statistically significant for SOEs with high and low state ownership. These results provide strong support for H4A.

Turning next to the valuation impact of the CSR regulation, we estimate the following model (derived from the residual income valuation model in Penman (2013)) to test the effect of mandatory CSR reporting on shareholder valuation (H4B):

$$(P/B)_{i,t} = \beta_t + \beta_i + \beta_1 TMT + \beta_2 POST + \beta_3 POST \times TMT + \beta_4 ROE_{i,t+1} + FIRM\,FE + \varepsilon_{i,t}. \tag{5}$$

In Equation (5), *P*/*B* is the market value of equity at the end of April in year $t+1$ to the book value of equity at the end of December in year t. We define $ROE_{i,t+1}$ as net income of year $t+1$ divided by book value of equity at the end of year t. If the primary effect of CSR reporting is one of wealth transfer from shareholders to other stakeholders, we should expect to see a decrease in the price-to-book ratio for treatment firms after the regulation. Table 8 presents these results.

We find in column (1) of panel A that the difference-in-differences effect on price-to-book ratio is significantly negative for the full sample ($\beta_3 = -0.9843$, $t = -5.59$, $p < 0.01$), indicating that the significant decrease in the price-to-book ratio for treatment firms is directly attributable to the CSR regulation. Columns (2) and (3) of panel A present results for the SOE and NSOE subsamples. The results for these subsamples are similar to those for the full sample, indicating that shareholders of both SOEs and NSOEs regard the heightened CSR activities as damaging shareholder value. Moreover, referring to panel B, the coefficient β_3 is negative and statistically significant for SOEs with high and low state ownership. Overall, these results provide strong support for H4B. These results are in line with there being a wealth transfer away from shareholders because they are bearing the cost.

5. Additional Analyses

5.1. Value Chain

Recent studies suggest that as firms focus on CSR, their value chain relationships may be affected for several reasons. Broadly speaking, corporate social responsibility encompasses suppliers and customers as they are important stakeholders of any firm. Indeed, the CSR guidelines issued by the Shenzhen

Table 8 Effects of CSR Disclosure on Shareholder Value

Panel A: SOEs vs. NSOEs

Dependent var. *P/B*		Column (1) *FULL* Coeff.	t value	Column (2) *SOE* Coeff.	t value	Column (3) *NSOE* Coeff.	t value
CONSTANT	β_0	2.9136	7.04***	−13.2665	−13.65***	3.2149	2.32**
TMT	β_1	−0.2161	−0.90	18.7427	15.54***	−0.1899	−0.13
POST	β_2	−0.9857	−7.07***	−1.1149	−8.20***	−1.0752	−3.63***
POST × TMT	β_3	−0.9843	−5.59***	−0.9797	−5.16***	−0.6827	−1.87*
ROE	β_4	4.3307	5.64***	2.9437	3.27***	4.9686	3.85***
FIRM FE		Yes		Yes		Yes	
Observations		6982		4562		2420	
Adj. R^2		0.360		0.482		0.349	

Panel B: Impact of state ownership

Dependent var. *P/B*		Column (1) High Coeff.	t value	Column (2) Low Coeff.	t value
CONSTANT	β_0	25.7048	15.42***	32.6212	24.72***
TMT	β_1	−23.7172	−16.93***	−26.7140	−16.53***
POST	β_2	−1.1406	−5.62***	−0.9072	−4.76***
POST × TMT	β_3	−0.7392	−2.76***	−1.2900	−4.16***
ROE	β_4	4.4543	2.75***	1.3035	1.12
FIRM FE		Yes		Yes	
Observations		2280		2282	
Adj. R^2		0.483		0.548	

Note: An SOE whose state ownership is above (below) the median percentage is classified into the “High” (“Low”) group. The *t*-statistics are adjusted for heteroscedasticity based on White’s (1980) approach. ***, **, and * denote two-tailed significance at the 1%, 5%, and 10% level, respectively.

Stock Exchange (SZSE 2008) requires firms affected by the regulation to disclose measures being taken to safeguard customer and supplier interests, which might heighten the global attention and scrutiny of CSR activities related to value chain relationships. Most SOEs and NSOEs are large firms that typically have dedicated networks of suppliers who presumably rely on them for their sustenance.[13] We would expect government-controlled SOEs help their suppliers by making quick payments and helping them manage their working capital better especially after the CSR mandate. The incentives of NSOEs are, however, less clear. When relationships with suppliers are market-driven and are working well, the CSR mandate per se should not have any effect for NSOEs. Hence, whether NSOEs reduce their demand for favorable credit terms from their suppliers following the mandate is an open question.

In a similar vein, recent studies have indicated that CSR can help a company build reputation and trust among customers and clients (Dutta and Singh 2013). Cheung and Pok (2019) advance and provide evidence in support of the “trust” view of CSR, which predicts that a CSR firm will engage in more trade credit *provision*. While government-controlled SOEs may change their behavior in line with the CSR mandate, the incentives of NSOEs are again less clear especially given that customer-relationships are shaped by market forces.

To address these value chain implications, we estimate Equation (1) using trade credit from suppliers and trade credit to customers (deflated by total assets) as the dependent variable, respectively.[14] Our results (untabulated) indicate that relative to control firms, treatment firms experience a significant decline in trade credit from suppliers in the post-regulation period. These results are attributable primarily to SOEs—we fail to detect any such effect for NSOEs. Moreover, we find that relative to control firms, treatment firms significantly increase trade credit to customers post-regulation. This result is driven primarily by NSOEs—we do not detect any such increase for SOEs. In addition, the extent of state ownership does not seem to play a role with respect to both supplier and customer relationships.[15]

Overall, we find some evidence that the CSR regulation has real effects in terms of improving value chain relationships. SOEs appear to demand less credit from suppliers following the regulation which suggests that as government-controlled entities, these firms reaffirm their commitment to CSR by paying greater attention to supplier welfare. Moreover, NSOEs appear to loosen credit terms on the customer side beyond what they would normally do absent regulation.

5.2. Growth
It is reasonable to argue that pre-regulation, high-growth firms are more likely to have ignored CSR to maintain growth. If so, one might expect our primary findings to be more pronounced for high-growth firms. To investigate this issue, we partition our sample into two equal-sized subsamples based on growth and replicate our analysis for each of these growth portfolios. As expected, untabulated results indicate that the real effects of the CSR regulation are uniformly more pronounced for high-growth portfolios relative to low-growth portfolios.

5.3. Financial Crisis
Although we use a difference-in-differences method to mitigate the effect of unobservable systematical factors on operations, our results may still be subject to the confounding event of the 2008 financial crisis. During the crisis, the Chinese government implemented a stimulus policy to provide credit capital to firms to mitigate the negative impact of the financial crisis. During the 2007–2010 period, sales for both treatment and control firms showed a steady upward trend, indicating that the stimulus was likely effective. In order to further exclude the effect of the financial crisis, we drop firm-years after 2009 from our analysis and our results remain qualitatively the same.

5.4. Falsification Test
To further strengthen the interpretation of the main results, we do a placebo test by replicating our analysis around a placebo period (Almeida et al. 2011, Bertrand et al. 2004), using 2007 as the "regulation" year. The falsification test can help us rule out alternative explanations caused by unobservable factors. If our results are caused by factors we have not identified, we should then obtain similar results even in this placebo period. However, the results indicate that our main findings are not attributable to any unidentified factors.

5.5. Sample of Control Firms
In our primary analysis, we exclude firms from our sample that are not subject to the disclosure mandate and that voluntarily disclose their CSR activities post-regulation. We exclude these firms because our focus is on the real effects triggered by the CSR regulation. We have no *a priori* reason to expect that the CSR regulation would induce these firms to *increase* the resources they devote to CSR activities post-regulation given that they do not have to face heightened global scrutiny. This argument holds especially for shareholder value-maximizing NSOEs. Nevertheless, as a robustness check, we include firms in our control group that who voluntarily issue CSR reports through the post-regulation period. Our results (untabulated) remain qualitatively the same.

5.6. Propensity Score Matching
In our analysis thus far, we have included all firms not subject to the disclosure mandate as control firms to examine the impact of regulation on affected firms. To the extent that treatment firms and control firms are associated with systematic pre-regulation differences in their profiles, any post-regulation differences that we detect may not be attributable to the regulation (Barrot 2015). Because the CSR regulation mandates disclosure only for firms meeting certain criteria, the treatment and control groups are non-random, and some systematic differences are bound to exist with any matching procedure based on observables. Nevertheless, to control for any identifiable differences, we employ a propensity score matching (PSM) procedure using pre-regulation period data. Untabulated results indicate that our inferences regarding the real effects of the CSR regulation remain qualitatively the same. Specifically, for SOEs subject to the regulation, the real effects manifest in the form of increases in core operating costs (COGS and total operating costs), and for NSOEs subject to the regulation, these effects appear in the form of increased CSR special items.

6. Conclusions

We use mandatory CSR reporting in China as an example to demonstrate the real effects of disclosure, thereby extending the real effects literature (Kanodia 2007, Kanodia and Sapra 2016) from a shareholder focus to a stakeholder focus. We document the real effects for both SOEs and NSOEs and, in particular, we find that they work through different channels. Direct CSR outlays for SOEs are spent on core operating activities whereas those for NSOEs are spent on non-core activities. Further, we document some real effects in the form of reductions in trade credit from suppliers for SOEs and increases in trade credit to customers for NSOEs. Finally, the effects of the mandatory CSR reporting on profitability and shareholder value are negative for both SOEs and NSOEs.

Admittedly, we have taken an initial step in examining how the CSR emphasis affects value chain relationships. An interesting avenue for future research is to delve deeper into this issue by identifying key suppliers and customers in the value chains of each firm and examining the underlying drivers of the real effects of CSR on their relative bargaining powers and efficiencies.

One potential limitation of our study is that we are not able to include CSR contributions that are non-monetary in nature. Some CSR items are non-monetary, for example, acres of greenery development, total number of employees trained, the amount of training hours, the amount of water saved, total greenhouse gas emissions and intensity. Other CSR disclosures involve generic statements of the CSR benefits, for example, community services provided to villages, without any mention of costs incurred. It is difficult to translate such disclosures into monetary units. Nevertheless, to get a rough estimate of the magnitude of CSR expenditures inclusive of such non-monetary items, we use the procedure used by Chen et al. (2018) and estimate the total CSR expenditures to be between ¥66.61 million (US$9.51 million) and ¥424.96 million (US$60.71 million).[16]

Non-monetary initiatives can be of different forms for different companies. By ignoring such non-monetary items, we are understating the magnitude of CSR initiatives both in the pre-regulation period and in the post-regulation period. Moreover, there is no way of identifying and quantifying how the market would view such items. This said, we have no reason to believe that excluding non-monetary items would bias our tests in any way. Clearly, excluding cases in which firms increase non-monetary CSR activities following the regulation works against us. By the same token, some firms may *cut* non-monetary items and instead increase monetary CSR expenditures. In this case, we are likely overestimating their response to the CSR regulation.

This paper looks at the real effects from firms' perspective. Because of the potential positive externalities and the deadweight loss, the net effect of CSR disclosure mandate on the social welfare could be either positive or negative. Because we do not have direct evidence on other stakeholders' responses to the CSR mandate and on externalities, we cannot directly measure the overall net effect. A potentially fruitful research agenda is to develop a theoretical framework of tracing the paths of the spillover effects, that is, how and to what extent non-shareholder stakeholders use the wealth transfer from corporate CSR activities in their own decision making and how it affects the social welfare and its components. Such a theoretical framework could help calibrate the parameters using real-world data. Overall, an examination of CSR from the perspectives of all stakeholders can further our understanding of the real effects of mandatory disclosure on social welfare.

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Data Availability Statement

Data used in this study are available from public sources.

Appendix A. Examples of Disclosures of CSR Expenditures

firm name	2007 (Pre-regulation)	2009 (Post-regulation)
Beiqi Foton Motor (SSE)	Donation to healthcare foundations: ¥1.5 million	Donation to hospitals: ¥4.3 millionDonation to youth development foundations: ¥1 millionEnvironmental treatment, workplace renovation, and greenery development: ¥1.8 millionSafety improvement: ¥40 million
Jiangxi Copper (SSE)	Environmental protection and energy savings: ¥9 million	Operating expenses of environmental protection facilities: ¥220 million
National Jiangshan Agrochemical & Chemicals (SSE)	Donation to governmental welfare programs: ¥0.56 million	Waste treatment: ¥22 millionSafety improvement: ¥18 millionOperating expenses of workplace safety: ¥8.7 million
Hebei Jinniu Energy Resources (SZSE)	Water pollution treatment: ¥3.2 million	Energy savings: ¥78.5 millionPollution prevention: ¥64.1 million
Shenzhen Agricultural Products (SZSE)	Water pollution treatment: ¥6 million	Water pollution treatment: ¥1 millionDonation to the Wenchuan earthquake victims: ¥4.3 million
Taiyuan Coal Gasification (SZSE)	Environmental protection and treatment: ¥50 million	Environmental protection and treatment: ¥37 millionSubsidies to low-income employees: ¥0.77 millionDonation to low-income students: ¥0.11 millionDonation to the Red Cross: ¥0.15 million

Appendix B. Variable Definitions

Dependent variables	
CSR_EXP	= CSR expenditures/total assets
CSR_CAP	= CSR capitalized expenditures/total assets
COGS	= cost of goods sold/sales
Total operating expenses	= (cost of goods sold + SG&A expenses)/sales
CSR_NOE	= CSR special items/sales
NOE	= non-core expenses/sales
AB_NOI	= industry-adjusted non-operating income/total assets (as in Chen and Yuan 2004)
ROA	= net income/beginning book value of total assets
ROE	= net income/beginning book value of equity
P/B	= market value of equity at the end of April in year $t + 1$ to book value of equity at the end of December in year t
Independent variables	
TMT	= a dummy variable that equals 1 if a firm was required to disclose a CSR report since 2008 and 0 otherwise
TMT^T	= a dummy variable that equals 1 for a treatment firm and 0 otherwise
POST	= a dummy variable that equals 1 for 2008 or thereafter and 0 otherwise
BEFORE	= a dummy variable that equals 1 if a firm belongs to the treatment group in 2007 and 0 otherwise
AFTER	= a dummy variable that equals 1 if a firm belongs to the treatment group in 2008 or thereafter and 0 otherwise
SOE	= an indicator variable that equals 1 if the firm is state-owned and 0 otherwise
SIZE	= the natural logarithm of total assets
LOSS	= an indicator variable that equals 1 if the net income before extraordinary items is negative, and 0 otherwise
CFO	= operating cash flow/total assets
LEV	= debt/total assets
GROWTH	= the sales growth rate
CROSSLIST	= an indicator variable that equals 1 if a firm issues shares to foreign investors and 0 otherwise
OWNERSHIP	= the percentage of shares held by the largest shareholder
AGE	= the natural logarithm of the number of years that a firm has been publicly traded
FIRM FE	=firm fixed effect

Notes

[1]In India, a regulation enacted in 2013 requires firms with a certain profile (based on profitability, net worth, and size) to spend at least 2% of their net income on CSR (Manchiraju and Rajgopal 2017).

[2]In a speech at the China NSOE Social Responsibilities Forum, Huang Mengfu, Vice Chairman of the Chinese People's Political Consultative Conference, the national political advisory body in China, believes that NSOEs are big players in education, healthcare, culture, environmental protection, and charities and that it is important for them to contribute to these causes and to be recognized for their contributions (Huang 2006).

[3]Many CSR expenditures disclosed by firms subject to the CSR regulation are capitalized initially and expensed later, which delays their impact on the income statement. An example is Yuguang Gold & Lead, whose environmental protection expenditure increased by ¥48 million in 2008 and ¥183 million in 2009. These expenditures were capitalized as constructions-in-progress and were converted into fixed assets that were subject to depreciation in 2010.

[4]We hand-collect data on CSR expenditures. For the pre-regulation period, the data source is the notes to the financial statements for treatment and control firms. For the post-regulation period, the data source is the mandatory stand-alone CSR reports for treatment firms and the notes to the financial statements for control firms.

[5]For example, Jiangxi Changyun (an NSOE) discloses in Note 51 to the 2012 financial statements that ¥1,328,633.80 spent on a flood control fund is treated as a special item.

[6]The literature also suggests that a CSR focus helps firms forge better value-chain relationships via building trust and reputation in their mutual dealings (Cheung and Pok 2019, Dutta and Singh 2013, Zhang et al. 2014).

[7]Chen et al. (2018) provide evidence that, for their full sample, non-core expenses of firms subject to the regulation do not increase, but they do not separately provide evidence for NSOEs.

[8]Some CSR initiatives are non-monetary in nature. We exclude non-monetary items in our empirical analysis. We defer a discussion of the nature of these non-monetary items and the consequent implications for our analysis in the concluding section.

[9]The SZSE mandated CSR reporting for companies listed in the Shenzhen 100 Index (SZSE 2008), and the SSE required CSR reporting for companies listed in the SSE Corporate Governance Sector, for those having shares listed abroad, and for financial services companies (SSE 2008).

[10]For this reason, we do not use the propensity score matching procedures for our main tests. As a robustness check, we report results using propensity score matching procedures in section 5.6. In addition, we are unable to use the regression-discontinuity design because we do not know the exact values of the cut-offs that separate treatment firms from control firms. For example, firms in the SSE Corporate Governance Sector are subject to the CSR mandate, and the membership is determined by 20 different indicators spanning many dimensions of corporate governance relating to operational compliance, shareholder behavior, characteristics of directors and senior management, and information disclosure. The exact values of the rating scores for these indicators are not disclosed, making it infeasible to use the regression-discontinuity design to identify control firms.

[11]We thank Dan Collins for his input with respect to the methodology.

[12]Because non-core items are not persistent, we do not perform a parallel trends test.

[13]As an example, SANY Heavy Industry (an NSOE) is a leading enterprise of high-end equipment with over 20 R&D centers and manufacturing bases all over the world. Its market value exceeded US$20 billion in 2019, making it a top three global construction machinery manufacturer. SANY is an absolute leader in its value chain according to its 2012 annual report.

[14] We also use sales and total assets less receivables and inventory as alternate deflators. Our results remain qualitatively the same.

[15] We also examine the *net* trade credit effect—we use the difference between trade credit from suppliers and trade credit to customers (deflated by total assets) as the dependent variable in our estimation. The untabulated results from this analysis reinforce our inferences from panels A to D.

[16] The ¥66.61 million is the *monetary* component of CSR spending disclosed by firms and excludes non-monetary items. Therefore, this amount represents the lower bound of CSR spending. The upper bound consists of two parts. First, the increase in total operating expenses of ¥401.31 million, which is the product of ¥14,700 million (sales) and 0.0273 (α_3 in column 1 of panel C in Table 5). Second, the increase in capitalized expenditures of ¥23.63 million, which is the product of ¥21,500 million (total assets) and 0.0011 (α_3 in column 1 of panel E in Table 6). The sum of the above two parts equals ¥424.96 million.

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